

1 Seaweeds are photosynthetic organisms. They contain chloroplast pigments, some of which are similar to land plants and some of which are different.

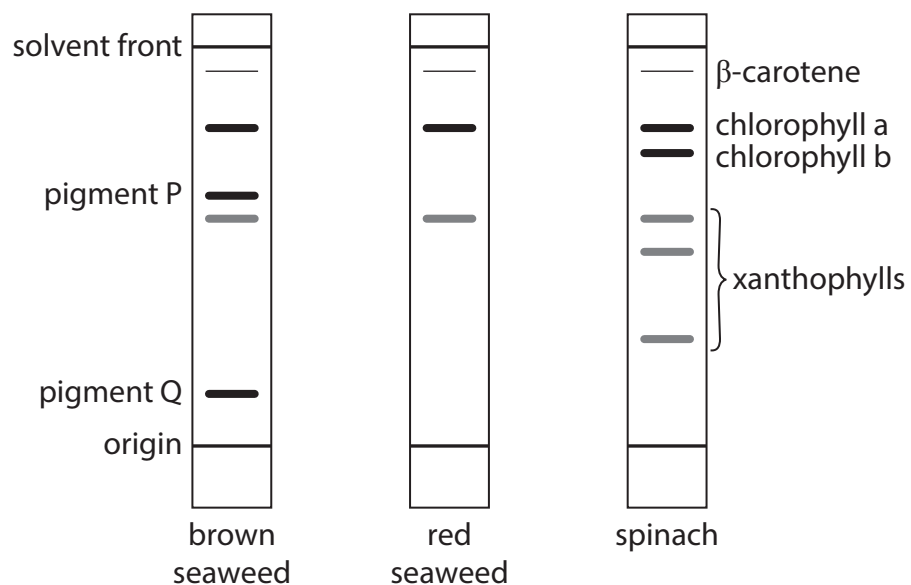
(a) Which process can be used to separate chloroplast pigments according to their solubility?

(1)

- ☒ **A** chromatography
- ☒ **B** dendrochronology
- ☒ **C** gel electrophoresis
- ☒ **D** polymerase chain reaction

(b) A student separated the chloroplast pigments found in brown seaweed, red seaweed and spinach using this method.

The diagram shows the results.



- (i) Compare and contrast the types of chloroplast pigment found in the seaweeds with those found in the spinach.

Use the information in the diagram to support your answer.

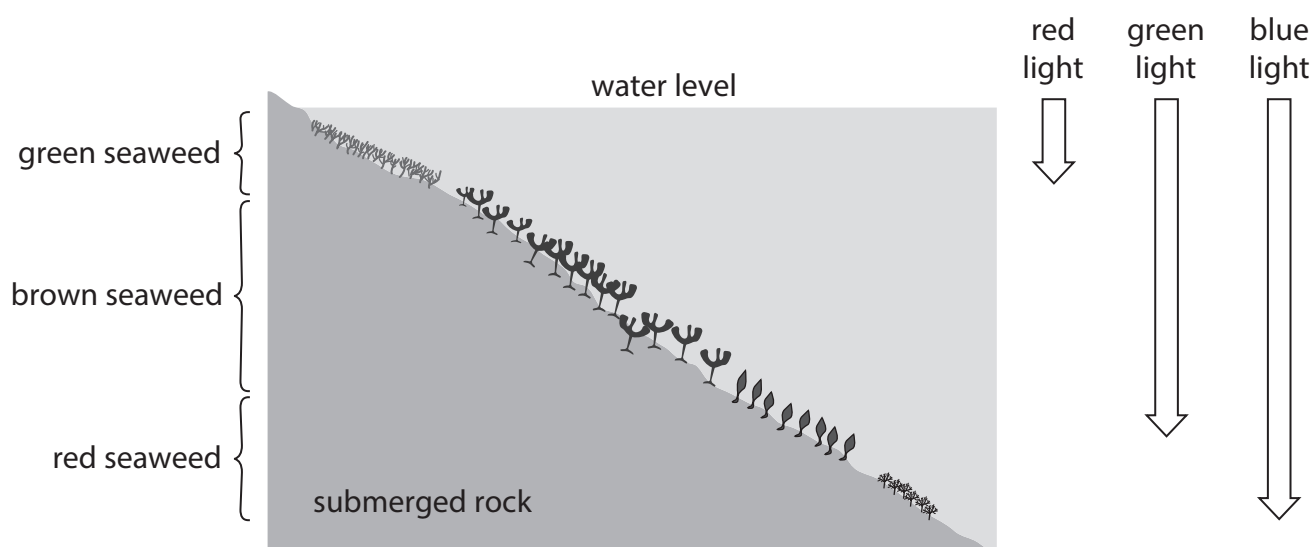
(3)

- (ii) Describe how the student could identify pigment P using this method.

(3)

- (c) The diagram shows the position of three different types of seaweed on rock covered in water.

The diagram also shows the depth to which red light, green light and blue light penetrate into the water.



- (i) Which row of the table shows the light that green seaweed can absorb?

(1)

	Green light	Red light
☐ A	can absorb	can absorb
☐ B	can absorb	cannot absorb
☐ C	cannot absorb	can absorb
☐ D	cannot absorb	cannot absorb

(ii) Explain the positions of the brown and red seaweeds on the submerged rock.

(2)

(Total for Question 1 = 10 marks)
